

agrees with G in the neighborhood of Π if $H \cap \Pi = G \cap \Pi$ and for each face f of H , there exists a corresponding face $\varphi(f)$ of G incident to exactly the same vertices of $G \cap \Pi$.

We point out that we can take $\varphi(f) = f$ in case f is contained in Π , since we assume that the embeddings of H and G coincide within Π . The nontrivial case arises when f is not contained in Π .

Lemma 13. *Let G be a 3-connected graph embedded in a fixed surface Σ with $\text{fw}(G) \geq 3$ and let (Π, Π^+) be a pair of nested surfaces with boundary contained in Σ and contoured by G . Then there exists a 3-connected surface minor H of $G \cap \Pi^+$ that agrees with G in the neighborhood of Π .*

Proof. We define H as the surface minor of $G \cap \Pi^+$ obtained by contracting each component of $(G \cap \Pi^+) - V(G \cap \Pi)$ into a single vertex. We obtain an embedding of H in $\widehat{\Pi}^+$ by modifying the canonical embedding of $G \cap \Pi^+$ in this surface. Specifically, we continuously contract the components of $(G \cap \Pi^+) - V(G \cap \Pi)$ and their enclosed faces into single points. This contraction step can be modeled by a continuous map $\sigma : \widehat{\Pi}^+ \rightarrow \widehat{\Pi}^+$, in such a way that the embedding of H is the image under σ of the embedding of $G \cap \Pi^+$. We choose σ to be injective outside of the components of $(G \cap \Pi^+) - V(G \cap \Pi)$, and the identity within Π . This can be done without loss of generality.

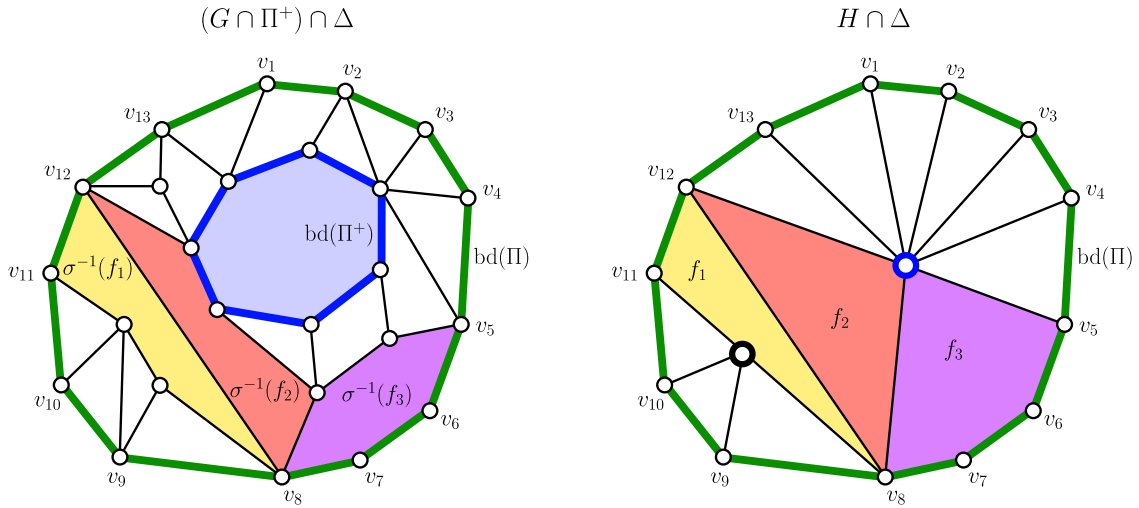


Figure 7: A graph $G \cap \Pi^+$ restricted to some disk $\Delta = \Delta_i$ which is a component of $\widehat{\Pi}^+ - \text{int}(\Pi)$ (left), and the corresponding minor H restricted to the same disk (right). The two larger vertices of H correspond to component vertices. The green and the blue cycle are cuffs (that is, boundary components) of Π and Π^+ respectively. The figure shows, for each $i \in [3]$, a face f_i of H contained in Δ along with and the corresponding face $\varphi(f_i) = \sigma^{-1}(f_i)$ of G .

We first argue that H agrees with G in the neighborhood of Π . Since σ is the identity within Π , we see immediately that $H \cap \Pi = G \cap \Pi$. Now consider some face f of H . We let $\varphi(f) := \sigma^{-1}(f)$ denote the corresponding face of G . We proceed to check that this is indeed a face of G , which is moreover incident to the same vertices of $G \cap \Pi$ as f .

If f is contained in Π , then we have $\sigma^{-1}(f) = f$ and there is nothing to check.

Therefore, we may assume that f is contained in $\widehat{\Pi}^+ - \Pi$. We observe that $\sigma^{-1}(f)$ is a face of G since we obtained H from $G \cap \Pi^+$ only by contracting edges. No edge or vertex was deleted. Because σ is the identity within Π , hence in particular on the boundary of Π , we see that $\sigma^{-1}(f)$ is incident to exactly the same vertices of $G \cap \Pi$ as f .